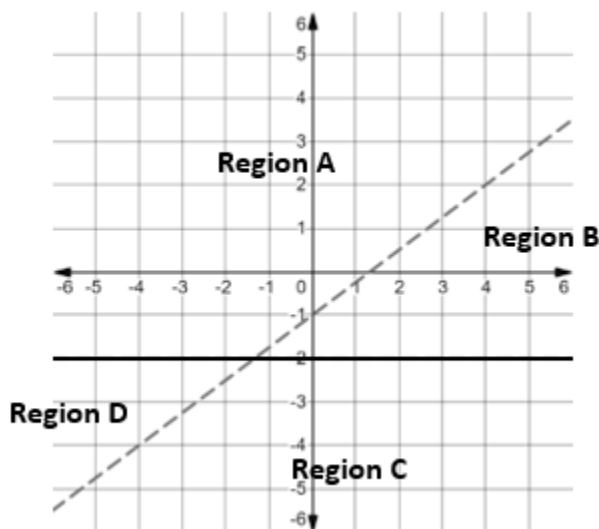


Algebra 1
Unit 6
Lesson 9 Homework

Name _____
Date _____
Period _____

Consider the system of inequalities, $\begin{cases} y \geq -2 \\ 3x - 4y < 4 \end{cases}$.

The boundary lines of both inequalities in the system are shown on the coordinate plane.



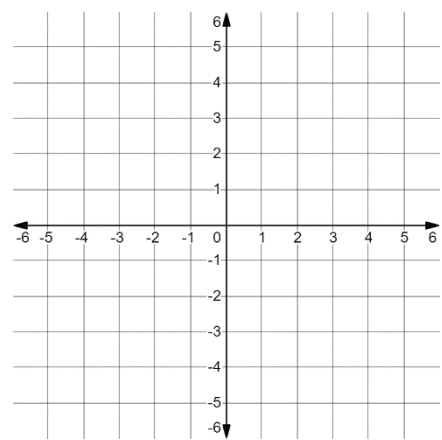
1. The boundary lines separate the plane into four regions, labeled A - D. Which region includes the solution set of the system of inequalities?
2. Shade the region that includes the solution set of the system of inequalities.
3. Select a test point in the shaded region of the solution set.
4. Substitute the x - and the y -values of the test point to show that the ordered pair satisfies both inequalities.

$y \geq -2$	$3x - 4y < 4$

5. Your friend says that the point $(3, -2)$ is a solution to the system of inequalities. Explain if you agree or not.

Consider the system of inequalities, $\begin{cases} y + 3 > 0.2(x + 5) \\ y - 4 \leq -2(x + 1) \end{cases}$.

6. Graph the boundary lines of the system.



Use substitution to determine whether the given ordered pairs are solutions to the system.

	$y + 3 > 0.2(x + 5)$	$y - 4 \leq -2(x + 1)$	Solution to the system?
7. $(-3,4)$			☐ Yes ☐ No
8. $(5,2)$			☐ Yes ☐ No

9. Complete the graph by shading the region included in the solution set of the system.

10. Graph the solution set of the system of inequalities: $\begin{cases} x \leq 2 \\ y > \frac{3}{4}x + 2 \end{cases}$.

